Chapter 6 • Trigonometry

Solved Exercise 6.1

1. State the quadrant and write one coterminal angle for each given angle.

(i) 65° lies between 0° and 90°, so it is in Quadrant I. A coterminal angle is 65°+360°=425°.
QI,425°

(ii) 135° lies between 90° and 180°, so it is in Quadrant II. A coterminal angle is 135°−360°=−225°.
QII,−225°

(iii) −40° lies between −90° and 0°, so it is in Quadrant IV. Add 360° to obtain 320°.
QIV,320°

(iv) 210° lies between 180° and 270°, so it is in Quadrant III. Subtract 360° to obtain −150°.
QIII,−150°

(v) −150° lies between −180° and −90°, so it is in Quadrant III. Add 360° to obtain 210°.
QIII,210°. Other coterminal answers differ by multiples of 360°.

2. Convert each decimal-degree measure to degrees, minutes and seconds.

(i) 123.456° = 123°+0.456(60′)
= 123°27.36′
= 123°27′+0.36(60″)
= 123°27′21.6″.

(ii) 58.7891° = 58°+0.7891(60′)
= 58°47.346′
= 58°47′+0.346(60″)
= 58°47′20.76″.

(iii) 90.5678° = 90°+0.5678(60′)
= 90°34.068′
= 90°34′+0.068(60″)
= 90°34′4.08″.

3. Convert each degrees-minutes-seconds measure to decimal degrees using $\mathbf{D+\frac{M}{60}\frac{+S}{3600}}$

(i) $65°32′15″ = 65+\frac{32}{60}\frac{+15}{3600}$
= 65.5375°.

(ii) $42°18′45″ = 42+\frac{18}{60}\frac{+45}{3600}$
= 42.3125°.

(iii) $78°45′36″ = 78+\frac{45}{60}\frac{+36}{3600}$
= 78.76°.

4. Convert each angle to radians using radians=degrees$\times\frac{\pi}{180}$

(i) $36°\times\frac{\pi}{180}=\frac{\pi}{5}$ rad.

(ii) $22.5°\times\frac{\pi}{180}=\frac{\pi}{8}$ rad.

(iii) $67.5°\times\frac{\pi}{180}=\frac{3\pi}{8}$ rad.

5. Convert each radian measure to degrees using degrees=radians$\times\frac{180}{\pi}$

(i) $(\pi/16)(180/\pi) = \frac{180}{16}$
= 11.25°.

(ii) (11π/5)(180/π) = 11(36)
= 396°.

(iii) (7π/6)(180/π) = 7(30)
= 210°.

6. Find the arc length and area of each sector.

(i) Given r=6 cm and $\theta=\frac{\pi}{3}$ rad: $\ell=r\theta=6(\frac{\pi}{3})=2\pi$ cm; $A=\frac{1}{2}r^2\theta=\frac{1}{2}(6^2)\frac{\pi}{3}=6\pi$ cm².
2π cm≈6.28 cm, 6π cm²≈18.85 cm²

(ii) Given $r=\frac{4.8}{\pi}$ cm and $\theta=\frac{5\pi}{6}$ rad: $\ell=(\frac{4.8}{\pi})\frac{5\pi}{6}=4$ cm.

studysapiens.com

For (ii), $A=\frac{1}{2}(4.8/\pi)^2(5\pi/6)=\frac{9.6}{\pi}$ cm²≈3.0558 cm².

4 cm, $\frac{9.6}{\pi}$ cm²≈3.06 cm².

7. A sector has angle 60° and radius 12 cm. Find its area and percentage of the circle.

$A = (\frac{60}{360})\pi(12^2)$
$= 24\pi$ cm²≈75.40 cm².
Percentage $= \frac{60}{360}\times 100$
= 16⅔%.
24π cm²≈75.4 cm²; 16.67%.

8. Find the percentage of a circle represented by a sector of angle $\frac{\pi}{8}$ radian.

Full angle=2π; percentage=$\frac{\frac{\pi}{8}}{2\pi}\times 100=\frac{100}{16}$
6.25%.

9. A radius-12 cm sector of angle 150° is bent into a cone. Find slant height and base radius R.

Sector radius becomes slant height: l=12 cm.
Arc=2π(12)(150/360)=10π cm. Thus 2πR=10π, so R=5 cm.
Slant height 12 cm; base radius 5 cm.

studysapiens.com